

Lab - Deploying a MySQL Database Container

To Begin with the Lab

Summary

This lab demonstrates how to deploy a **MySQL database** inside an **Azure Container Instance (ACI)**. Initially, the deployment from Docker Hub fails due to Docker Hub's anonymous pull rate limit. As a workaround, the MySQL image is first pulled to the local machine, pushed to **Azure Container Registry (ACR)**, and then deployed successfully from ACR. Finally, the MySQL container is accessed to create a database, user, tables, and insert data.

Understanding

A **MySQL container** packages the MySQL database engine with all its required dependencies into a Docker image. Instead of installing MySQL on a virtual machine, you can run it directly as a container. Azure Container Instances provides the compute infrastructure automatically, so you only need to provide the container image and configuration.

Docker Hub hosts official MySQL images, but anonymous pull requests are subject to rate limits. To avoid this issue, the image can be downloaded locally, tagged, pushed to Azure Container Registry, and then deployed from ACR. MySQL uses environment variables (such as `MYSQL_ROOT_PASSWORD`) during the first startup to initialize the database securely.

Example:

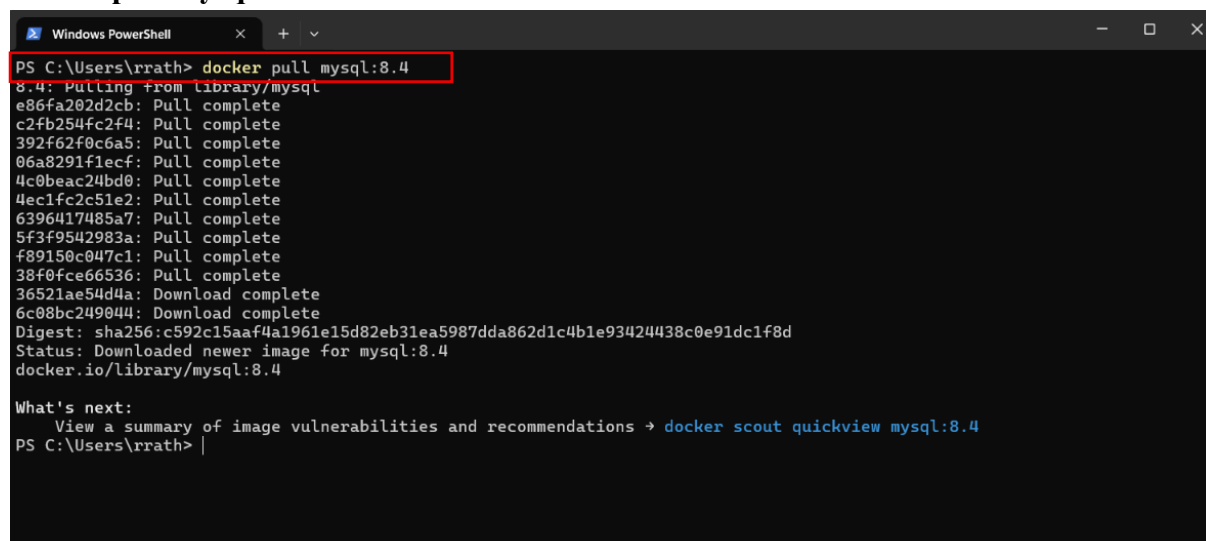
- Without Docker → Install MySQL manually on a VM.
- With Docker → Run MySQL inside a container using the official MySQL image.
- With Azure Container Instances → Azure manages the infrastructure while MySQL runs inside the container.

Lab Steps

Step 1: Pull MySQL Image Locally

- Open the powershell.

docker pull mysql:8.4

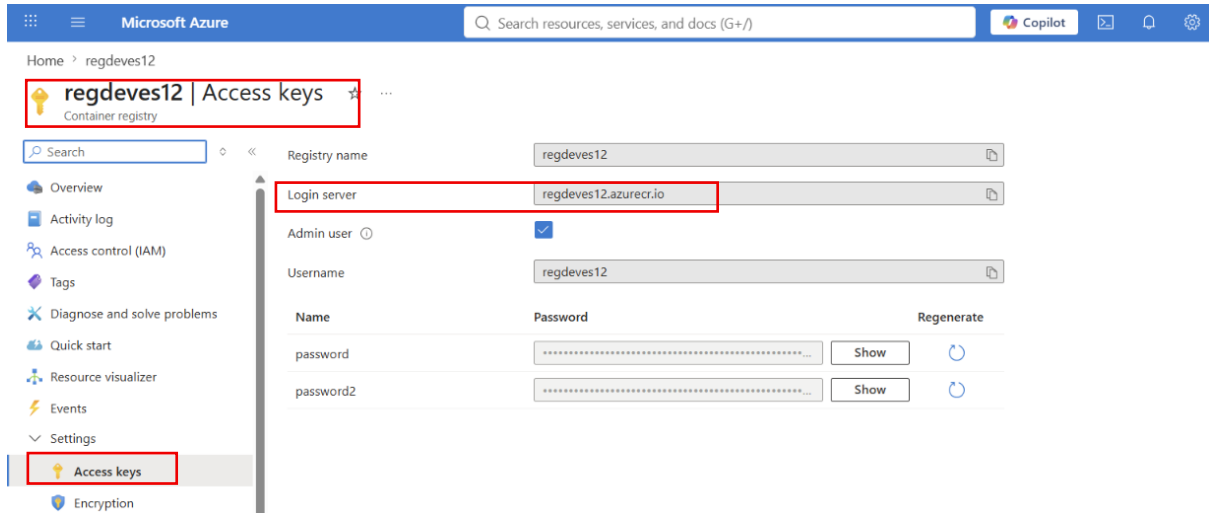


```
Windows PowerShell
PS C:\Users\rrath> docker pull mysql:8.4
8.4: Pulling from library/mysql
e86fa202d2cb: Pull complete
c2fb254fc2f4: Pull complete
392f62f0c6a5: Pull complete
06a8291f1ecf: Pull complete
4c0beac24bd0: Pull complete
4ec1fc2c51e2: Pull complete
6396417485a7: Pull complete
5f3f9542983a: Pull complete
f89150c047c1: Pull complete
38f0fce66536: Pull complete
36521ae54d4a: Download complete
6c08bc249044: Download complete
Digest: sha256:c592c15aaf4a1961e15d82eb31ea5987dda862d1c4b1e93424438c0e91dc1f8d
Status: Downloaded newer image for mysql:8.4
docker.io/library/mysql:8.4

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview mysql:8.4
PS C:\Users\rrath> |
```

Step 2: Tag the Image for Azure Container Registry

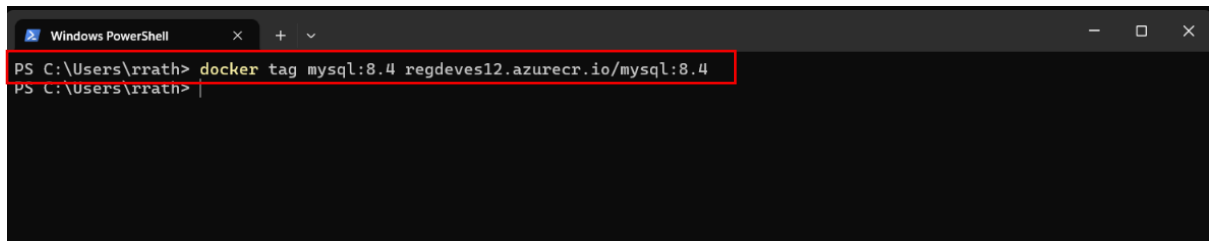
- Go to Azure container Registry.
- Navigate to Access keys.
- Copy the Login server.



docker tag mysql:8.4 <ACR-Login-Server>/mysql:8.4

Example:

docker tag mysql:8.4 regdev12.azurecr.io/mysql:8.4

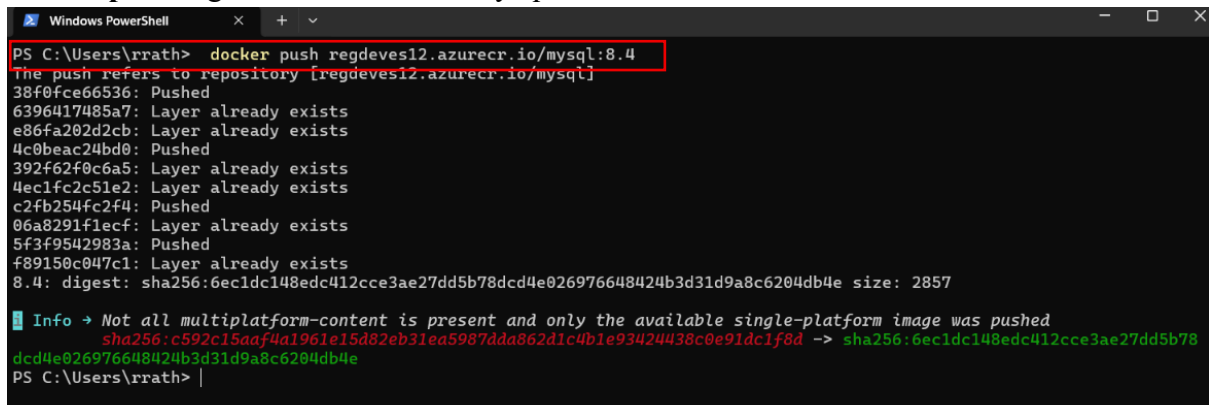


Step 3: Push Image to Azure Container Registry

docker push <ACR-Login-Server>/mysql:8.4

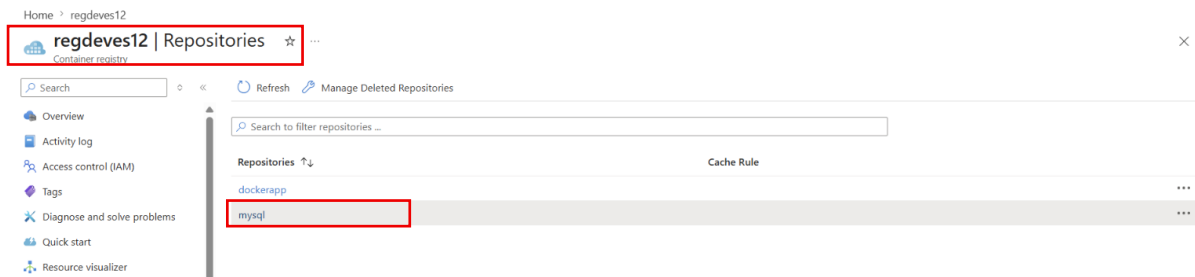
Example:

docker push regdev12.azurecr.io/mysql:8.4



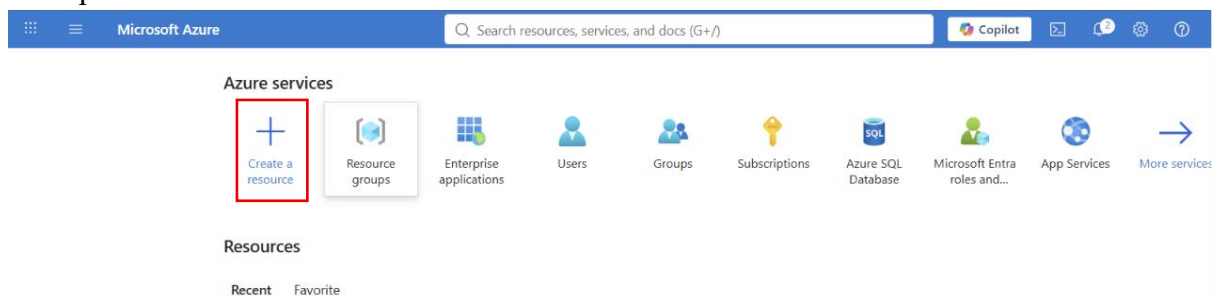
Step 4: Verify Image in Azure Container Registry

- Open Azure Container Registry.
- Select Repositories.
- Verify that the mysql:8.4 image is available.

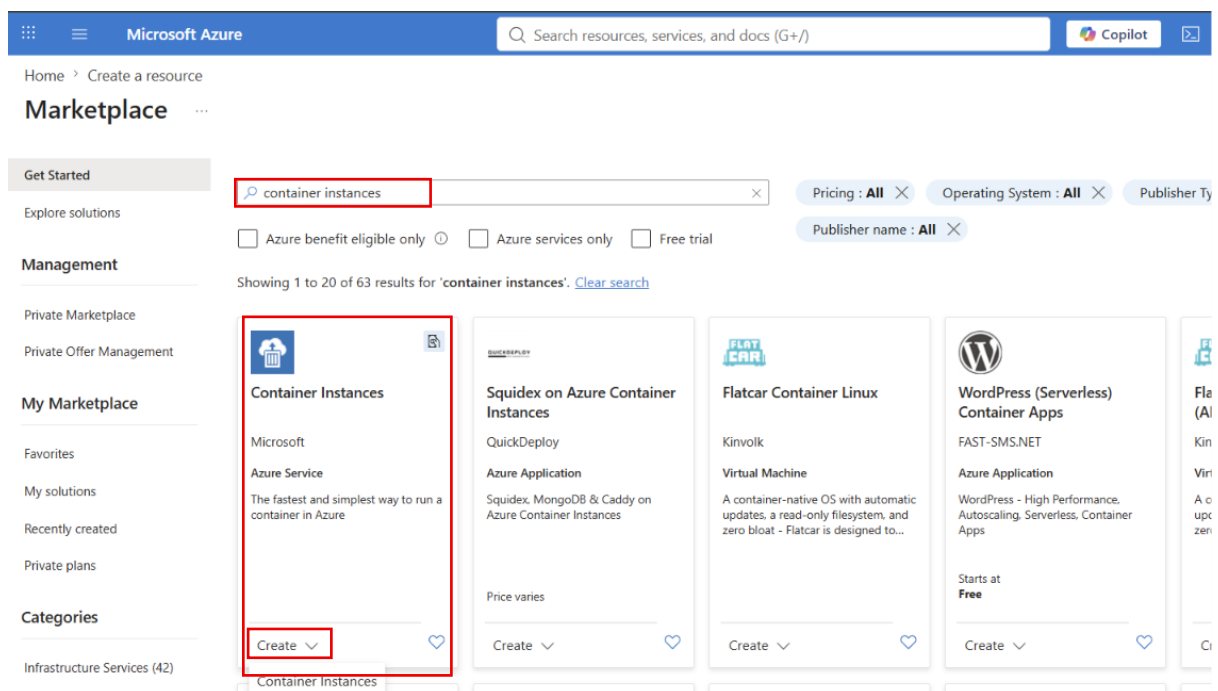


Step 5: Open Azure Container Instances

- Sign in to the Azure Portal.
- Open **Create a Resource**.



- Search for **Container Instances**.
- Click **Create**.



Step 6: Configure Basic Settings

- Select the existing **Resource Group**.
- Enter a **Container Name**.
- Select the **Region** (Example: East US).

Create container instance ...

Basics Networking Monitoring Advanced Tags Review + create

Azure Container Instances (ACI) allows you to quickly and easily run containers on Azure without managing servers or having to learn new tools. ACI offers per-second billing to minimize the cost of running containers on the cloud.

[Learn more about Azure Container Instances](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Visual Studio Enterprise Subscription

Resource group * ⓘ Webapp

[Create new](#)

Container details

Container name * ⓘ mysql-aci

Region * ⓘ (US) East US

Availability zones (Preview) ⓘ None

SKU Standard

Step 7: Select Docker Hub Image

- Choose **Image Source** → **Other Registry**.
- Specify the image:
mysql:8.4
- Keep **OS Type** as **Linux**.

Create container instance ...

Region * ⓘ (US) East US

Availability zones (Preview) ⓘ None

SKU Standard

Image source * ⓘ

☐ Quickstart images

☐ Azure Container Registry

☒ Other registry

⚠ Please be aware that Docker Hub has recently introduced a pull rate limit on Docker images. When specifying an image from the Docker Hub registry, this may impact the creation of your container instance. [Learn more](#)

Run with Azure Spot discount ⓘ

☐ Spot containers are not available in the selected region. [Learn more](#)

Image type * ⓘ

☒ Public ☐ Private

Image * ⓘ mysql:8.4

ⓘ If not specified, Docker Hub will be used for the container registry and the latest version of the image will be pulled.

OS type *

☒ Linux ☐ Windows

ⓘ This selection must match the OS of the image chosen above.

Size * ⓘ

1 vcpu, 1.5 GiB memory, 0 gpus

[Change size](#)

Step 8: Configure Networking

- Enable **Public IP** (for the lab).
- Expose **TCP Port 3306** (default MySQL port).

Home > Create a resource > Marketplace

Create container instance ...

Basics **Networking** Monitoring Advanced Tags Review + create

Choose between three networking options for your container instance:

- **'Public'** will create a public IP address for your container instance.
- **'Private'** will allow you to choose a new or existing virtual network for your container instance.
- **'None'** will not create either a public IP or virtual network. You will still be able to access your container logs using the command line.

Networking type ☒ Public ☐ Private ☐ None

DNS name label ⓘ

DNS name label scope reuse ⓘ Any reuse (unsecure) ▼

Ports ⓘ

Ports	Ports protocol
3306 ✓	TCP

Step 9: Configure Environment Variable

Add the following environment variable:

Variable	Value
MYSQL_ROOT_PASSWORD	mysql@123

- Mark the password as **Secure**.

Home > Create a resource > Marketplace

Create container instance ...

Basics Networking Monitoring **Advanced** Tags Review + create

Configure additional container properties and variables.

Restart policy ⓘ On failure ▼

Environment variables

Mark as secure	Key	Value
No ▼	MYSQL_ROOT_PASSWORD ✓	mysql@123 ✓
No ▼		

Command override ⓘ Example: ["/bin/bash", "-c", "echo hello; sleep 100000"]

Key management ⓘ ☒ Microsoft-managed keys (MMK) ☐ Customer-managed keys (CMK)

Step 10: Deploy the Container

- Click **Review + Create**.
- Click **Create**.

Create container instance ...

✓ Validation passed

Basics Networking Monitoring Advanced Tags Review + create

Basics

Subscription	Visual Studio Enterprise Subscription
Resource group	Webapp
Region	East US
Container name	mysql-aci
SKU	Standard
Image type	Public
Image	mysql:8.4
OS type	Linux
Memory (GiB)	1.5
Number of CPU cores	1
GPU type (preview)	None
GPU count	0

Networking

Networking type	Public
Ports	3306 (TCP)
DNS name label scope reuse	Any reuse (unsecure)

Create

< Previous

Next >

[Download a template for automation](#)

- **Wait until deployment completes.**

Home

 **Microsoft.ContainerInstances-20260722134548** | Overview ...

Deployment

x <<  Delete  Cancel  Redeploy  Download  Refresh

 Overview

 Inputs

 Outputs

 Template

✓ **Your deployment is complete**

 Deployment name : Microsoft.ContainerInstances-20260722134548
Subscription : [Visual Studio Enterprise Subscription](#)
Resource group : [Webapp](#)

Start time : 7/22/2026, 1:48:22 PM
Correlation ID : 6fea07de-ed41-4569-8dae-43b4a16b6adc

> Deployment details

✓ Next steps

[Go to resource](#)

Give feedback

 [Tell us about your experience with deployment](#)

Step 11: Verify Container Status

- Open the deployed **Container Instance**.
- Navigate to **Containers**.
- Verify the container state is: Running

Home > Microsoft.ContainerInstances-20260722134548 | Overview > mysql-aci

mysql-aci | Containers

Container instances

Search Refresh Give feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Containers**
- Identity
- Properties
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- Monitoring
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- Help

1 container and 0 init containers

Name	Image	State	Previous state	Start time	Restart count
mysql-aci	mysql:8.4	Running	-	2026-07-22T08:19:02.874Z	0

Events Properties Logs Connect

Display time zone Local time UTC

Name	Type	First timestamp	Last timestamp	Message	Count
Started	Normal	7/22/2026, 01:49:02 PM GMT+5...	7/22/2026, 01:49:02 PM GMT+5...	Started container	1
Pulled	Normal	7/22/2026, 01:48:53 PM GMT+5...	7/22/2026, 01:48:53 PM GMT+5...	Successfully pulled image 'mys...	1
Pulling	Normal	7/22/2026, 01:48:30 PM GMT+5...	7/22/2026, 01:48:30 PM GMT+5...	pulling image 'mysql@sha2566...	1

Step 12: View Container Logs

- Open Containers → Logs.
- Verify the message:
- MySQL daemon is ready for connections.

This confirms that the MySQL server has started successfully.

Home > Microsoft.ContainerInstances-20260722134548 | Overview > mysql-aci

mysql-aci | Containers

Container instances

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1 container and 0 init containers

Name	Image	State	Previous state	Start time	Restart count
mysql-aci	mysql:8.4	Running	-	2026-07-22T08:19:02.874Z	0

```

2026-07-22 08:19:15+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-07-22T08:19:16.4332012 Z [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.4.10).
2026-07-22T08:19:19.7598192 Z [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.4.10) MySQL Community Server - GPL.
2026-07-22T08:19:19.7682532 Z [System] [MY-015016] [Server] MySQL Server - end.
2026-07-22 08:19:20+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-07-22 08:19:20+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

2026-07-22T08:19:20.452643Z Z [System] [MY-015015] [Server] MySQL Server - start.
2026-07-22T08:19:20.633756Z Z [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.4.10) starting as process 19
2026-07-22T08:19:20.638759Z Z [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-07-22T08:19:21.383348Z Z [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-07-22T08:19:21.700693Z Z [Warning] [MY-010060] [Server] CA certificate ca.pem is self signed.
2026-07-22T08:19:21.700806Z Z [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-07-22T08:19:21.728944Z Z [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS user
s. Consider choosing a different directory.
2026-07-22T08:19:21.778124Z Z [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock
2026-07-22T08:19:21.778600Z Z [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.10' socket: '/var/run/mysqld/mysqld.sock' port: 330
6 MySQL Community Server - GPL.
  
```

Step 13: Connect to the MySQL Container

- Open Containers → Connect.
- Click on the Connect.

Home > Microsoft.ContainerInstances-20260722134548 | Overview > mysql-aci

mysql-aci | Containers

Container instances

Search Refresh Give feedback

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Containers

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1 container and 0 init containers

Name	Image	State	Previous state	Start time	Restart count
mysql-aci	mysql:8.4	Running	-	2026-07-22T08:19:02.874Z	0

Events Properties Logs **Connect**

Choose Start Up Command

☒ /bin/bash

☐ /bin/sh

☐ Custom

Connect

- Start the MySQL client:

```
mysql -u root -p
```

- Enter the **root** password.

Events Properties Logs **Connect**

```

bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 118
Server version: 8.4.10 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
  
```

Step 14: Create a Database

```
CREATE DATABASE CourseDB;
```

```
mysql> CREATE DATABASE CourseDB;
Query OK, 1 row affected (0.02 sec)
```

Step 15: Create a User

```
CREATE USER 'courseuser'@'%' IDENTIFIED BY 'Password123!';
```

```
mysql> CREATE USER 'courseuser'@'%' IDENTIFIED BY 'Password123!';
Query OK, 0 rows affected (0.03 sec)
```

Step 16: Grant Permissions

```
GRANT ALL PRIVILEGES ON CourseDB.* TO 'courseuser'@'%';
FLUSH PRIVILEGES;
```



```
mysql> GRANT ALL PRIVILEGES ON CourseDB.* TO 'courseuser'@'%';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)
```

Step 17: Use the Database

USE CourseDB;

```
mysql> USE CourseDB;
Database changed
```

Step 18: Create a Table

```
CREATE TABLE CourseOrders (
    OrderID INT PRIMARY KEY AUTO_INCREMENT,
    CourseName VARCHAR(100),
    StudentName VARCHAR(100),
    Amount DECIMAL(10,2)
);
```

```
mysql> CREATE TABLE CourseOrders (
->     OrderID INT PRIMARY KEY AUTO_INCREMENT,
->     CourseName VARCHAR(100),
->     StudentName VARCHAR(100),
->     Amount DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.09 sec)
```

Step 19: Insert Data

```
INSERT INTO CourseOrders (CourseName, StudentName, Amount)
VALUES
('Webdeveloping', 'Tushar', 199.99),
('Software-Engineering', 'Ayush', 149.99);
```

```
mysql> INSERT INTO CourseOrders (CourseName, StudentName, Amount)
-> VALUES
-> ('Webdeveloping', 'Tushar', 199.99),
-> ('Software-Engineering', 'Ayush', 149.99);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql>
```

Step 20: View Data

SELECT * FROM CourseOrders;

```
mysql> SELECT * FROM CourseOrders;
+-----+-----+-----+-----+
| OrderID | CourseName | StudentName | Amount |
+-----+-----+-----+-----+
| 1 | Webdeveloping | Tushar | 199.99 |
| 2 | Software-Engineering | Ayush | 149.99 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql>
```